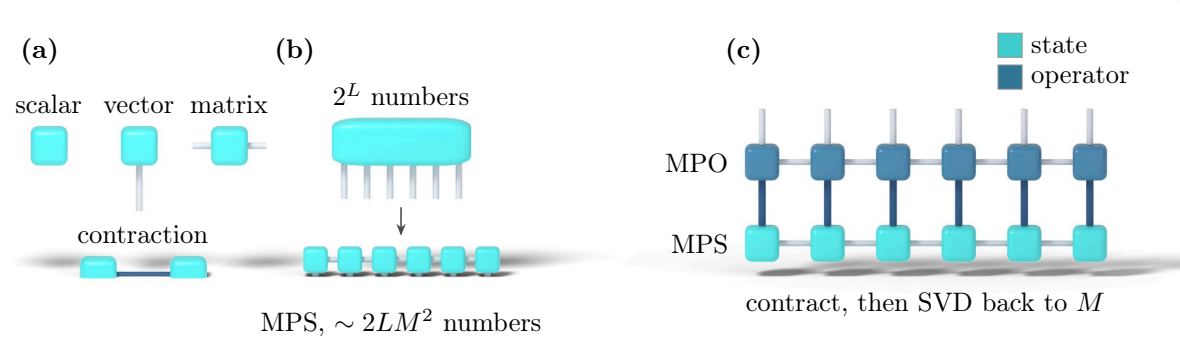

(a)
scalar
vector
matrix
contraction
(b)
2^L numbers
MPS, $\sim 2LM^2$ numbers
(c)
state
operator
MPO
MPS
contract, then SVD back to M